The Application Model of Generative AI in the Development of Graduate Students' Research Capabilities

Abstract

Driven by the profound transformation of global research paradigms through generative artificial intelligence (AI), this study explores how generative AI can enhance the research capabilities of graduate students. Addressing challenges such as inefficient literature reviews, insufficient innovation in experimental design, and the slow adaptation of training systems to new technologies, the research proposes a systematic application model. The study elaborates on empirical practices across the entire research workflow, including literature pro-cessing, experimental design, paper writing, and ethical education. It also pre-sents strategies to tackle challenges like excessive technical dependence, data security threats, and outdated ethical standards. The study concludes with future research directions, including cross-cultural comparisons of research capa-bilities, multi-technological synergy, long-term impact tracking, and dy-namic adaptation to emerging disciplines. It aims to provide theoretical support and practical references for the deep integration of "AI + education."